

Power Budget

Team Number: 201

Project Name: Garden Buddy

Team Member Name: Rylee Wirt

Version: 02

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	+1.8 - 5.1V	1	100	100	mA
	WT-1205 Buzzer	WT-1205	+5V	1	50	50	mA
	MCP6004 Quad Op Amp	MCP6004-I/P	+1.8 - 5.5V	1	1	1	mA
	Red LED indicator	WP710A10SRD/J4	N/A	1	2	2	mA
	LM7805 Linear Regulator	LM7805CT	+7 - 25V	1	5	5	mA
	Regulator quiescent	N/A	+5.0V	1	5	5	mA
+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	N/A	N/A	N/A	0	N/A	N/A	mA
	Subtotal					0	mA
	Safety Margin					25%	
	Total Current Required on +12V Rail					0	mA
Regulator	N/A	N/A	N/A	0	N/A	N/A	mA
Total Remaining Current Available on +12V Rail						0	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano Board	DM164150	+1.8 - 5.1V	1	100	100	mA
	WT-1205 Buzzer	WT-1205	+5V	1	50	50	mA
	MCP6004 Op Amp	MCP6004-I/P	+1.8 - 5.5V	1	1	1	mA
	Red LED Indicator	WP710A10SRD/J4	+1.8 - 2.2V	1	2	2	mA
	Regulator/Wiring Allowance	N/A	5.0V rail	1	5	5	mA
	Subtotal					158	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					197.5	mA
Regulator	LM7805 Linear Voltage Regulator	LM7805	+7 - 25V	1	1500	1500	mA
Total Remaining Current Available on +5V Rail						1302.5	mA

+3.3V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	N/A	N/A	N/A	0	N/A	N/A	mA
	Subtotal					0	mA
	Safety Margin					25%	
	Total Current Required on +3.3V Rail					0	mA
Regulator	N/A	N/A	N/A	0	N/A	N/A	mA
Total Remaining Current Available on 3.3V Rail						0	mA

<i>External Power Source 1</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	5V Wall Adapter (Regulated, Certified)	Generic UL/CE	100 - 240VAC	+5.0V	2000	10000	mA
Power Rails Connected to External Power Source 1	N/A	N/A	N/A	N/A	N/A	N/A	mA
	+5V Regulator	LM7805	+7 - 25V	1	1500	1500	mA
	N/A	N/A	N/A	N/A	N/A	N/A	mA
Total Remaining Current Available on External Power Source 1						8500	mA

<i>External Power Source 2</i>	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 2 Selection	N/A	N/A	N/A	N/A	N/A	N/A	mA
Power Rails Connected to External Power Source 2	N/A	N/A	N/A	N/A	N/A	N/A	mA
Total Remaining Current Available on External Power Source 2						#VALUE!	mA

Calculated Battery Life							
	Component Name	Part Number	Supply Voltage Range		Capacity (mAh)	Required By Regulators	
	N/A	N/A	N/A	N/A	N/A	N/A	
Battery Life						#VALUE!	hours

Notes:
 External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
 If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit